

Method:

For this lab I used simplex tableau to solve the linear programming problem.

Approach:

First, `build_tableau` creates the initial tableau matrix using the input arrays.

Then, `simplex_pivot_column` selects the largest positive coefficient, and `simplex_pivot_row` uses the returned column to select the pivot row by calculating the ratio for each row and selecting the smallest.

Finally, `simplex_pivot` performs the Gauss-jordan elimination by dividing the pivot row to make the pivot entry 1, then subtracting out the pivot row from the rest of the rows to make the rest of the pivot column entries 0.

Reasoning:

This solution efficiently solves the linear programming problem by repeatedly improving the objective function through simplex iterations until the optimal solution is reached.

Reflection:

For this lab I used the starter file with the built in driver to check correctness.